

SHAMBHAVI JAISWAL

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RESEARCH INTEREST

Cosmology and astrophysics; early Universe and large-scale structure; observational and computational astrophysics; transient and time-domain astronomy; galaxy evolution; statistical and Bayesian inference; machine learning for astrophysical data analysis; computational modelling and simulations.

RESEARCH EXPERIENCE

Project Associate I - Algorithm Development for CMB Analysis

Guide : [Dr. Santanu Das](#), Co-Guide : [Dr. Krishna Mohan Parattu](#) | IIT Mandi | Oct 2024 – March 2025

- Developed realistic multi-frequency **Planck-like CMB temperature and polarization (E-mode)** simulations incorporating asymmetric beam effects, scan patterns, Galactic foregrounds, and anisotropic instrumental noise.
- Built an end-to-end data-analysis pipeline for simulating observations and modeling foreground contamination (dust, synchrotron, and instrumental noise).
- Developed a **GAN/U-Net framework** for simultaneous CMB temperature and polarization reconstruction, foreground removal, and beam deconvolution.

Cosmological Parameter Estimation using SCoPE

Guide : [Dr. Santanu Das](#), Co-Guide : [Dr. Krishna Mohan Parattu](#)

- Contributed to the development and refinement of **SCoPE (Slick Cosmological Parameter Estimator)**, a parallelized MCMC framework for Bayesian cosmological parameter estimation.
- Developing a **machine-learning emulator** to accelerate cosmological parameter inference by approximating Boltzmann solver predictions within the SCoPE framework.
- Integrated **Planck likelihoods** through the **clik** interface and validated the MCMC engine for Λ CDM and extended cosmological models.
- Optimized and deployed the framework on Linux/HPC environments, including MPI-based parallel execution and scientific library integration (GSL, LAPACK, CFITSIO)

Project Associate I - Machine Learning to Uncover Megastructures in Transit Light Curves

Supervisor: [Dr. Vikram Khaire](#) | IIT Tirupati | Apr 2025 – July 2025

- Developed a deep-learning framework to reconstruct **2D opacity maps** of transiting objects from stellar light curves.
- Generated realistic synthetic training datasets using **BATMAN**, **EightBitTransit**, Bézier-curve models, and quadratic limb-darkening.
- Investigated inverse reconstruction of anomalous transit signatures, including debris structures, exocomets, and potential technosignatures.

M.Sc. Dissertation - X-ray Spectral Analysis of a Seyfert I Galaxy

Guide : [Dr. Main Pal](#), Co-Guide : [Dr. Krishna Mohan Parattu](#)

- Performed broadband X-ray spectral analysis using SAS, HEASoft, and XSPEC.
- Conducted time-resolved spectroscopy and multi-wavelength correlation analysis to investigate spectral variability.
- Studied the physical mechanisms governing X-ray emission and accretion processes in a Narrow-Line Seyfert I galaxy.

PUBLICATIONS & PREPRINTS

Obasho M, Shambhavi Jaiswal, Santanu Das, Krishna Mohan Parattu

Deep Learning for CMB Foreground Removal and Beam Deconvolution: A U-Net GAN Approach

Accepted in *Journal of Cosmology and Astroparticle Physics (JCAP)*; arXiv:2509.00139 [astro-ph.IM] (2025)

Vikram Khair, Gitika Shukla, Shambhavi Jaiswal, Obasho M

Machine Learning to Predict the Shadow Image of the Transiting Object from its Transit Light Curve

Manuscript in preparation

PROJECTS

Planetary Radius Distribution in Binary Systems

- Investigated the planetary radius distribution and radius gap in planets hosted by binary stars.
- Studied how the radius distribution changes with binary separation and examined the possible impact of uncertainty in identifying the planet-hosting star.
- Explored how these observational uncertainties can affect the inferred properties and population-level distribution of planets in binary systems.

Field-Level Inference for Weak Lensing

ICTS-TIFR Program : The Concurrence of Mega-Surveys

July–August 2026

- Worked on Bayesian field-level inference for weak-lensing data, focusing on reconstructing the underlying dark-matter distribution directly from galaxy survey observations.
- Explored the use of differentiable cosmological simulations, Hamiltonian Monte Carlo (HMC), Bayesian inference, and machine learning for extracting cosmological information directly from weak-lensing fields.
- Studied field-level reconstruction methods and compared reconstructed mass maps obtained from weak-lensing observations.

TECHNICAL SKILLS

Programming Languages: Python, C/C++ , Matlab, Fortran, R.

Scientific Computing: Linux, HPC Computing, MPI, OpenMP, SLURM, Bash, Git

Data Analysis & Scientific Python: NumPy, SciPy, Pandas, Matplotlib, scikit-learn

Machine Learning: PyTorch, TensorFlow, Convolutional Neural Networks (CNNs), U-Net, GANs, Physics-Informed Neural Networks (PINNs)

Statistical Method: Bayesian Inference, MCMC, Statistical Data Analysis, Hypothesis Testing, Numerical Methods

Cosmology & Astrophysics: CAMB, CLASS, HEALPix/healpy, Astropy, XSPEC, SAS, HEASoft

High-Performance Computing: MPI, GSL, LAPACK, CFITSIO

CONFERENCES, WORKSHOPS & SCHOOLS

The Concurrence of Mega-Surveys, ICTS-TIFR, Bengaluru, India

27 Jul – 7 Aug 2026

Exploring Exoplanets: The Journey to Discover New Frontiers and Habitable Worlds, ICTS-TIFR, Bengaluru, India

16 – 27 Mar 2026

Indo-Italian Workshop in Physics 2026, IIT Delhi, India

25 – 28 Feb 2026

Science Day Workshop, IIT Mandi, India

28 Feb 2024

EDUCATION

Indian Institute of Technology Mandi, H.P.

Master of Science in Physics

2022 - 2024

Deen Dayal Upadhyaya Gorakhpur University, U.P.

Bachelor of Science in Physics, Chemistry & Mathematics

2018 - 2021

ADDITIONAL INFORMATION

Hobbies & Interests: Trekking, mountain cycling, Nature exploration, travelling, badminton, chess, cooking

Community Engagement: Volunteered as a science tutor for underprivileged children.

REFERENCES

- Dr. Krishna Mohan Parattu
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- Dr. Vikram Khaire
Assistant Professor, Department of Physics, Indian Institute of Technology Tirupati
Email: vikramk@iittp.ac.in